

Advanced Wearable Biosensors and Flexible Elastomeric Platforms: A Review on Non-Invasive Molecular Monitoring, Materials Architecture, and AIoT Convergence

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Abstract

Traditional healthcare models are undergoing a profound paradigm shift toward personalized, preventive, and predictive medicine. Wearable flexible chemical sensors and biosensors have emerged as the cornerstone of this transformation, enabling continuous, real-time, and non-invasive monitoring of physiological biomarkers from alternative biofluids such as eccrine sweat, saliva, tears, and interstitial fluid (ISF). This review provides a comprehensive analysis of the mechanical, chemical, and electronic configurations that govern state-of-the-art epidermal diagnostic platforms. We evaluate the fundamental transduction mechanisms—including amperometric, potentiometric, voltammetric, and optical modalities—and examine the materials science underlying compliant substrates (plastics, textiles, and elastomers) and stretchable interconnects. Furthermore, we explore advanced soft microfluidic routing networks, skin-interfaced ultrasound profiling, and the integration of these sensors into wireless body area networks (bodyNET) optimized by Artificial Intelligence (AI) and the Internet of Things (IoT). Finally, we address key translation challenges including biocompatibility, biofouling, and signal artifact calibration, outlining future paths toward autonomous, self-powered clinical diagnostics suitable for high-impact commercial and research ecosystems.

1. Introduction: The Paradigm Shift in Personalized Healthcare

The escalating economic burden of global healthcare, combined with an rapidly aging world population, has generated an urgent demand for medical technologies capable of monitoring patient health status outside conventional clinical settings. The Internet of Things (IoT) and high-speed wireless networks are poised to revolutionize future healthcare ecosystems, with wearable devices driving a multi-trillion-dollar economic impact worldwide. Historically, wearable fitness monitors have relied strictly on biophysical or electrophysiological physical parameters, such as heart rate, blood pressure, skin temperature, respiration rate, and motion, alongside electrophysiological tracking like electrocardiography (ECG), electroencephalography (EEG), and electromyography (EMG). While these biophysical indicators offer crucial insights into cardiovascular and neuromuscular functions, they completely lack the capacity to provide direct, molecular-level information regarding underlying dynamic biochemical and metabolic processes.

To bridge this diagnostic gap, wearable chemical sensors and biosensors have emerged as a highly attractive class of skin-interfaced devices. By shifting the diagnostic paradigm away from invasive, episodic blood analysis toward the continuous, *in situ* analysis of non-invasive biofluids—namely eccrine sweat, saliva, tears, interstitial fluid, wound exudate, and exhaled breath—these systems offer a pristine, stress-free window into human physiology. Molecular analysis of these alternative fluids facilitates remote monitoring of chronic diseases, early biomarker detection for genome mutations, real-time physiological profiling for elite athletes, and exhaustion tracking for high-risk operators. This review systematically explores the architectural components of wearable biosensors, details their analytical transduction modes, compares the mechanical properties of compliant substrates, investigates temporal fluidic routing, and charts

their convergence with artificial intelligence to realize autonomous, closed-loop point-of-care medical monitoring.

2. Structural Component Architecture of Biosensors

A biosensor is defined as an integrated analytical device that converts a biological or molecular recognition event directly into an analytically useful, measurable electrical or optical signal. As established by Professor Leland C. Clark Jnr (widely recognized as the father of modern biosensors), the functional architecture of any advanced biosensing system relies fundamentally on the intimate coupling of three distinct operational elements :

1. **The Bioreceptor (Biological Recognition Element):** A highly specific, immobilized biological entity (such as an enzyme, antibody, oligonucleotide DNA/RNA sequence, hormone-sensitive transcription factor, cell-wall receptor, or whole microorganism) that selectively interacts with or binds to the target analyte of interest, eliminating chemical interference from surrounding matrix components.
2. **The Physicochemical Transducer:** An interfacial element in direct contact with the bioreceptor layer. It systematically translates the physical, mass, thermal, optical, or electrochemical modifications resulting from the biomolecular interaction into a proportional electronic or photonic output.
3. **The Signal Processing Unit (Detector/Microprocessor):** An electronic framework responsible for signal conditioning, amplification, and algorithmic processing, converting raw current, voltage, or frequency shifts into standardized concentration units displayed on a digital interface or transmitted wirelessly to remote storage.



Figure 1: Schematic taxonomy of wearable and flexible chemical sensors for non-invasive health monitoring, illustrating the transition from invasive blood sampling to alternative biofluids (sweat, saliva, tears, breath, ISF, and wound exudate) alongside their primary electrochemical and optical transduction principles.

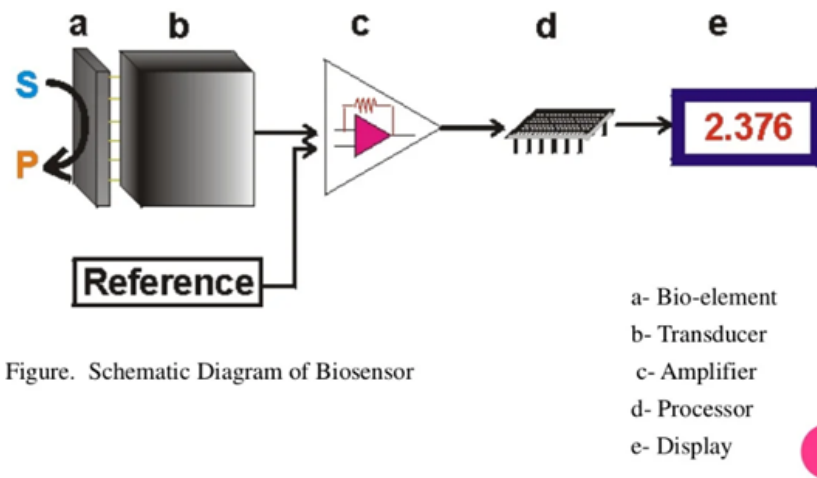


Figure. Schematic Diagram of Biosensor

Figure 2: The universal operational pipeline of an electrochemical/optical biosensor. (a) The bio-element immobilization layer reacting with target analytes (S substrate, P product), (b) the transducer core with an embedded reference cell, (c) signal amplification circuitry, (d) the microprocessing array, and (e) the linearized quantitative digital read-out display.

3. Advanced Physicochemical Transduction Modalities

The classification of wearable chemical sensors is predominantly dictated by the physical nature of their biotransducer. Electroanalytical and optical strategies represent the most reliable methods for translating biological dynamics into real-time digital metrics.

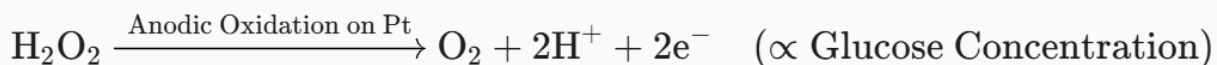
3.1. Electrochemical Sensors

Electrochemical transduction configurations are highly favored for wearable platforms due to their outstanding sensitivity, low power consumption, minimal instrumental complexity, and seamless compatibility with miniaturized, printed circuit architectures.

3.1.1. Amperometric Biosensors

Amperometry involves applying a constant, predetermined potential between a functionalized working electrode and a reference electrode, subsequently measuring the resulting current generated by the oxidation or reduction of electroactive species involved in a specific biochemical redox reaction. Enzymatic amperometric biosensors are universally utilized for tracking metabolic substrates such as glucose, lactate, ethanol, and uric acid. In these systems, specific oxidase enzymes—such as glucose oxidase (GOD), lactate oxidase (LOx), and urate oxidase—are immobilized on the working electrode surface to catalyze target metabolite reactions. For example, in a glucose amperometric sensor, glucose reacts with GOD to yield gluconic acid and hydrogen peroxide (H_2O_2) while consuming dissolved oxygen. The local

oxidation of H₂O₂ at a noble metal working electrode (such as platinum or gold) yields a direct electrical current that scales linearly with the substrate concentration :



3.1.2. Potentiometric Sensors

Potentiometry is implemented as a zero-current technique that monitors the electrical potential difference established across a permselective membrane between an ion-selective working electrode (ISE) and an isolated reference electrode. This technique is exceptionally well-suited for quantifying electrolyte shifts (Na⁺, K⁺, Ca²⁺, Cl⁻, and pH) critical to hydration tracking and acid-base balancing. The analytical correlation between ion activity in the biofluid and the registered voltage output is governed precisely by the classical Nernst Equation :

$$E = E^0 + \frac{RT}{zF} \ln \left(\frac{A_{\text{outside}}}{A_{\text{inside}}} \right) = \text{Constant} + S \log(A)$$

Where E represents the measured cell potential, E⁰ is the standard formal potential, R is the universal gas constant, T is the absolute temperature, z represents the ionic valency, F is the Faraday constant, and S denotes the theoretical sensitivity slope (59.16 mV/decade for monovalent ions like sodium or potassium at room temperature). Solid-state ion-to-electron transducers utilizing conductive polymers (e.g., polyaniline) or nanostructured carbon networks have enabled the fabrication of highly stable, miniaturized potentiometric arrays on flexible substrates.

3.1.3. Voltammetric Sensors

Voltammetry drives information regarding one or multiple chemical analytes dynamically by scanning the applied potential across a predefined range and registering the resulting current signature as a function of that varied voltage. Anodic Stripping Voltammetry (ASV) is highly effective for heavy metal analysis within trace body fluids. By integrating effective electrochemical preconcentration steps (electrodeposition) followed by a selective linear stripping oxidation scan, microelectrode arrays composed of gold, bismuth, or carbon hybrids can simultaneously quantify ultra-low concentrations (parts-per-billion or parts-per-trillion scales) of mineral contaminants and heavy metals, including zinc (Zn), cadmium (Cd), lead (Pb), copper (Cu), and mercury (Hg).

3.2. Optical, Colorimetric, and Photometric Transducers

Optical biosensors bypass the operational complexities of on-skin power infrastructure by leveraging colorimetric or fluorometric chemical assays. These devices incorporate color-responsive reagents, pH dyes, or responsive ionogels embedded within a soft matrix. When exposed to target analytes (e.g., chloride ions, lactate, or glucose), the reagents undergo distinct, irreversible or reversible changes in light absorption, wavelength, or emission intensity within the visible range. Quantitative results are generated via digital image processing using smartphones or personal mobile devices, matching captured RGB (Red-Green-Blue) data profiles against pre-calibrated colorimetric matrices, offering an elegant solution for resource-limited point-of-care environments.

4. Materials Science: Compliant Substrates and Stretchable Interconnects

Conventional microelectronics rely on rigid materials such as silicon, glass, and fiberglass composites, which exhibit high Young's moduli (10^9 to 10^{12} Pa) and brittle physical mechanics. Conversely, the human epidermis is soft (10^5 to 10^6 Pa Young's moduli), highly curvilinear, and continuously subject to multidirectional deformation, stretching, and shear stress. This mechanical mismatch induces severe motion artifacts, localized skin irritation, delamination, and device failure. Overcoming these limitations requires advanced mechanics design and compliant materials with low Young's moduli.

4.1. Evaluation of Flexible and Stretchable Materials

Three primary classes of soft materials serve as modern biosensor substrates: plastics, textiles, and elastomers.

1. **Flexible Plastics:** Thin-film polymeric substrates such as polyethylene terephthalate (PET), polyimide (PI), polyester (PE), and temporary inkjet-printed transfer tattoo papers possess low flexural rigidity and high chemical inertness. They are highly compatible with additive patterning techniques, including screen printing, gravure printing, and inkjet deposition. However, their poor gas/moisture breathability and inability to stretch restricts their application to brief monitoring intervals or highly miniaturized, decentralized structural nodes.
2. **Breathable Textiles:** Woven fabric meshes, natural cottons, and synthetic fibers (e.g., nylon, spandex) offer supreme breathability and comfort. They provide excellent surface area for sweat collection and can naturally interface across expansive bodily contours. Fabrication challenges involve addressing surface roughness and fluidic wicking porosity, which can lead to poor adhesion of printed conductive tracks. This is mitigated by applying thin elastomeric priming base layers prior to functional material patterning.
3. **Soft Elastomers:** Low-modulus silicone compositions (e.g., polydimethylsiloxane [PDMS], Ecoflex, Solaris) and advanced hydrogels represent highly compliant substrate options. Their stress-strain behaviors can be custom-tailored to perfectly mimic the

characteristic "J-shaped" mechanical response of human skin, ensuring tight, waterproof sealing and long-term, motion-robust epidermal adhesion.

4.2. Stretchable Interconnect Architectures

To maintain robust electrical connectivity between rigid integrated circuit (IC) processing blocks and soft epidermal chemical sensors across high mechanical strain, five stretchable interconnect modalities are utilized :

- **Serpentine Interconnects:** Thin, narrow metallic films or wires patterned into micro-machined, spring-like serpentine loops that geometrically accommodate applied stress by flexing out-of-plane rather than stretching structural lattices.
- **Conductive Elastomers:** Composite matrices formed by embedding conductive nanomaterials (such as carbon nanotubes [CNTs], graphene flakes, or silver nanoparticles) uniformly within a soft elastomeric backbone to achieve high ionic/electronic conductivity along with large mechanical stretchability.
- **Liquid-Metal Interconnects:** Microfluidic channels filled with liquid-phase alloys (such as Galinstan, a eutectic gallium-indium-tin blend) that maintain metallic conductivity and fluidic flexibility under extreme multidirectional deformation.
- **Conductive Inks:** Nanoparticle suspensions tuned to precise printable viscosities, designed to maintain functional network pathways upon curing, even when deposited onto highly flexible or stretchable surfaces.
- **Textile-Integrated Conductive Elements:** Functionalized garments featuring thin metallic wires woven directly into textile structures, or fabrics uniformly coated with conductive polymers (e.g., PEDOT:PSS), allowing seamless biomechanical compliance and washability.

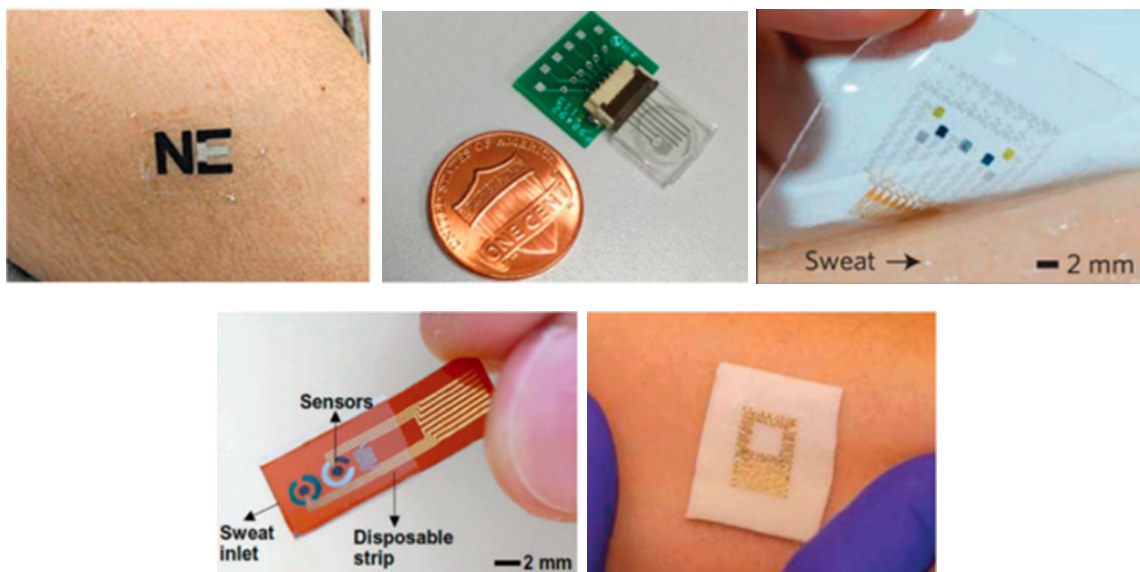


Figure 3: Structural manifestations of skin-interfaced biochemical sensor arrays. (a) Temporary-transfer tattoo electrochemical patch optimized for *in situ* lactate and alcohol profiling, (b) a miniaturized multiplexed sensor board resting adjacent to a copper coin for scale, (c) a

stretchable microelectrode array showcasing flexible mechanics under peel deformation, (d) a microfluidic disposable strip indicating the localized sweat inlet, and (e) a serpentine bilayer mesh consisting of gold-doped graphene forming a highly stable electrical transfer interface.

5. Microfluidic Sampling Networks and Analyte Profiles

The collection and routing of biofluids represent critical operational tasks for accurate real-time diagnostics. Human skin contains approximately 200 eccrine sweat glands per square centimeter across most anatomical regions, which secrete sweat at highly unpredictable and irregular rates influenced by physical exercise, thermal stress, age, and metabolic state.

5.1. Soft Microfluidic Systems

Advanced epidermal microfluidics constructed from soft silicones interface directly with the skin surface, trapping raw sweat immediately as it emerges from the dermal pores. These channels isolate the sample from the atmosphere, eliminating diagnostic errors caused by sample evaporation, environmental particulate contamination, and surface oils or cellular debris.

Through the integration of passive capillary design principles developed by the lab-on-a-chip community, modern wearable microfluidic patches utilize precise hydrophobic passive valves and localized superabsorbent polymer chambers (e.g., crosslinked sodium polyacrylate networks). These systems route incoming sweat flows sequentially through isolated reaction reservoirs containing distinct chemical or enzymatic assay reagents. This chronological sample separation completely eliminates analyte cross-talk, controls fluidic directionality solely via the natural secretory pressure of the sweat glands, and facilitates precise time-dependent mapping of biomarker fluctuations. For environments where active physical exercise is missing, active sweat excretion can be induced locally within minutes via iontophoresis, utilizing mild electrical current to drive pilocarpine or agonist agents into the targeted dermal tissue.

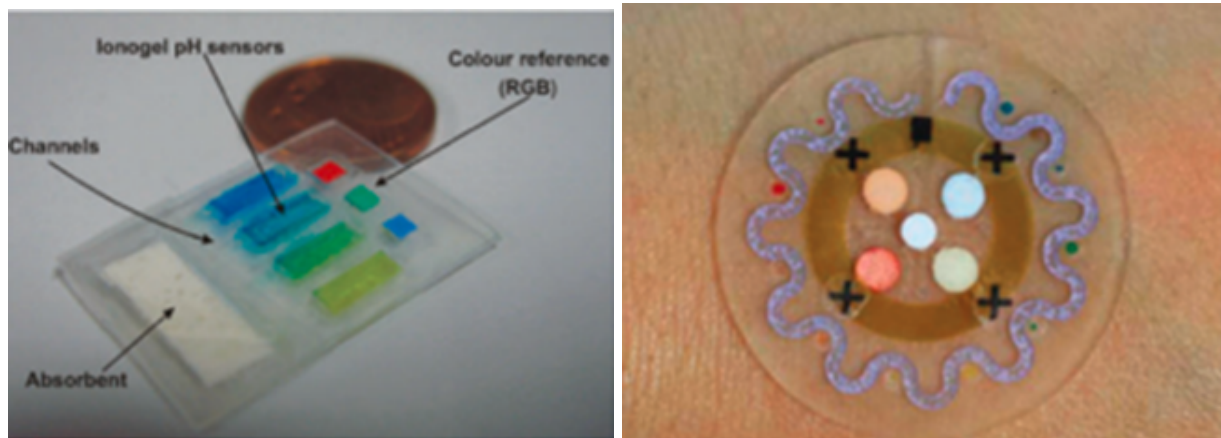


Figure 4: Advanced soft microfluidic systems for automated sweat collection and routing. Left: A multi-channel elastomeric microfluidic platform drawing sweat through distinct ionogel networks containing colorimetric pH indicators. Right: A circular, skin-conformable microfluidic patch with concentric serpentine tracks and multiple colorimetric assay spots optimized for synchronized mapping of sweat rate, total fluid loss, chloride concentrations, lactate, and pH.

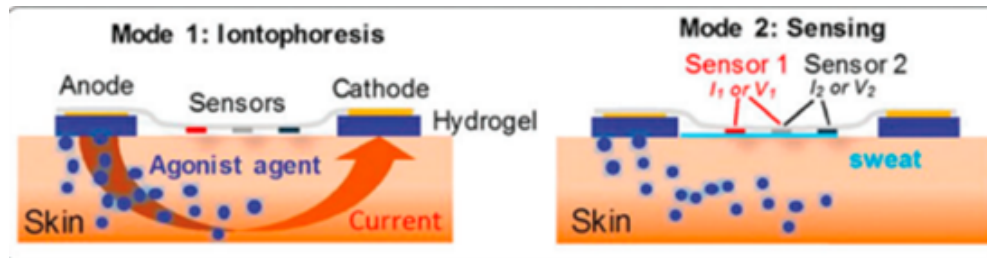


Figure 5: Cross-sectional operational overview of an integrated epidermal iontophoresis and sensing system. Mode 1 illustrates active delivery of a charged agonist agent (pilocarpine) from hydrogel layers into the skin glands via mild current. Mode 2 outlines subsequent real-time analyte quantification by integrated electrochemical working sensors as sweat flows across the interface.

5.2. Comprehensive Diagnostic Biomarker Directory

The clinical significance of non-invasive monitoring platforms relies on the biochemical correlation between target analytes within alternative biofluids and established physiological health indices.

- **Electrolyte Dynamics (Na^+ , K^+ , Cl^-):** Fluctuations in electrolyte levels provide directly quantified parameters regarding systemic dehydration, physical fatigue, and electrolyte imbalance. Elevated sodium levels indicate severe hydration loss, while elevated potassium and calcium dynamics can flag muscular stress, kidney issues, or metabolic conditions like Alzheimer's and diabetes.
- **Metabolite Biomarkers (Glucose, Lactate, Uric Acid):** Salivary and sweat glucose concentrations track systemic blood-glucose shifts non-invasively for diabetes management. Lactate tracking serves as a metric for aerobic/anaerobic thresholds in sports rehabilitation, while uric acid profiles indicate kidney performance and gout risks. Moreover, sweat ethanol concentrations maintain a direct linear correlation with blood-alcohol levels, enabling continuous sobriety monitoring.

6. Hormonal Tracking and Skin-Interfaced Ultrasonic Profiling

Translating wearable platforms into deeper clinical domains requires moving past basic metabolites to target trace hormones and deep tissue structural imaging.

6.1. Real-Time Hormonal Monitoring

Monitoring hormones like cortisol and estradiol is highly valuable due to their direct regulatory control over mental stress adaptation, vascular blood pressure, reproductive rhythms, and psychological disorders. However, real-time tracking has historically posed severe biological and engineering challenges due to the low concentrations of hormones in alternative biofluids and the slow regeneration rates of affinity surfaces.

To address these hurdles, innovative interdisciplinary strategies merge microbial genetics, protein engineering, and advanced optoelectronics into smart bracelets. For instance, a novel estradiol-sensing patch incorporates highly specific hormone-sensitive transcription factors isolated from microbial organisms, functionalized onto biopolymer nanoparticles (dendrimers and polypeptide arrays) alongside fluorescent molecular beacons (MB). The transduction mechanism relies on Fluorescence Resonance Energy Transfer (FRET). In the baseline state, the transcription protein forces the fluorescent entities into a tight complex, completely quenching light emission via FRET. Upon binding estradiol present in sweat, the molecular beacon undergoes a conformational separation, terminating FRET and releasing an intense fluorescent optical signal captured by an onboard optoelectronic interface.

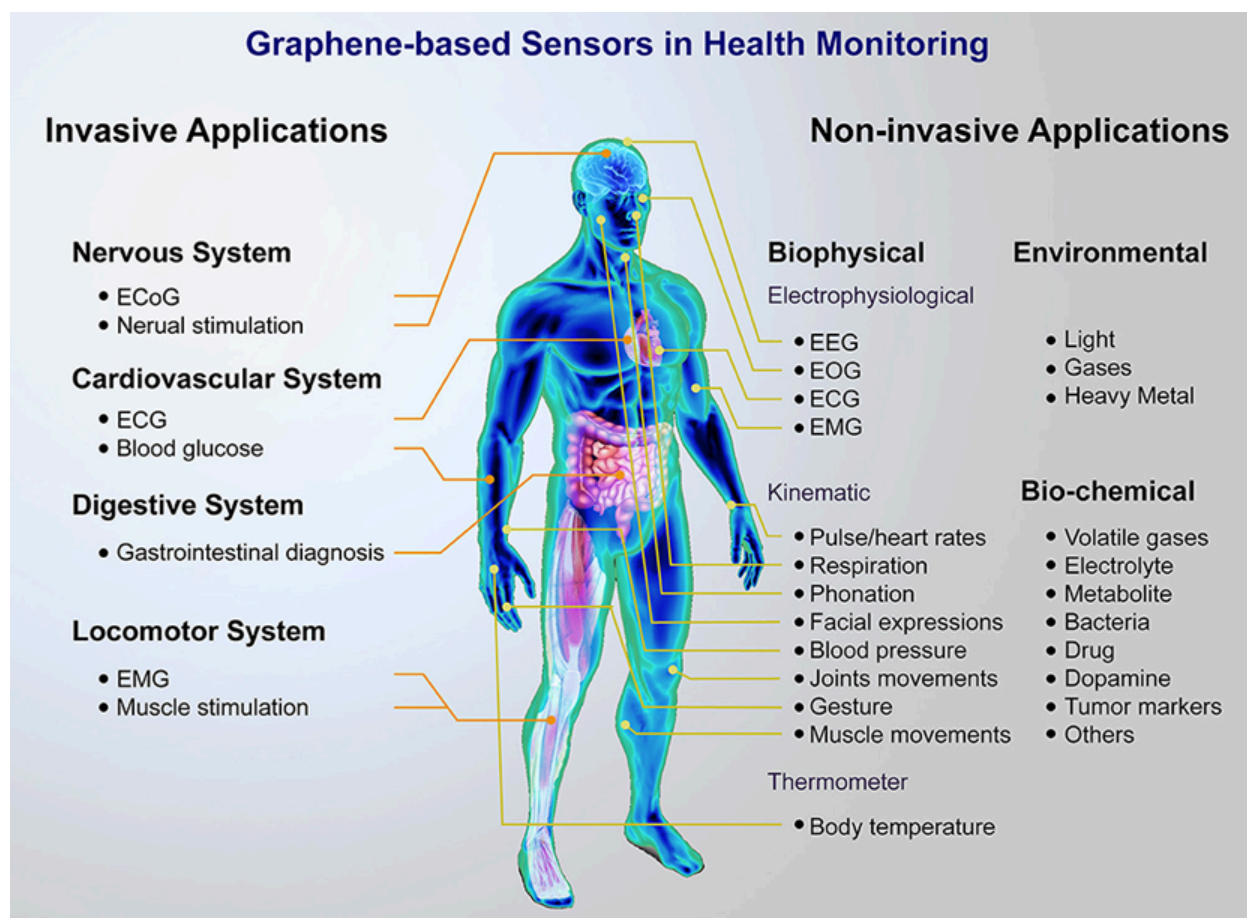


Figure 6: Diagnostic landscape of graphene-hybrid architectures in health monitoring. The diagram maps invasive neural/cardiovascular tracking alongside non-invasive epidermal modalities, detailing biophysical (EEG, ECG, EMG, kinetics) and biochemical (gases, electrolytes, metabolites, hormones, drugs) sensing vectors.

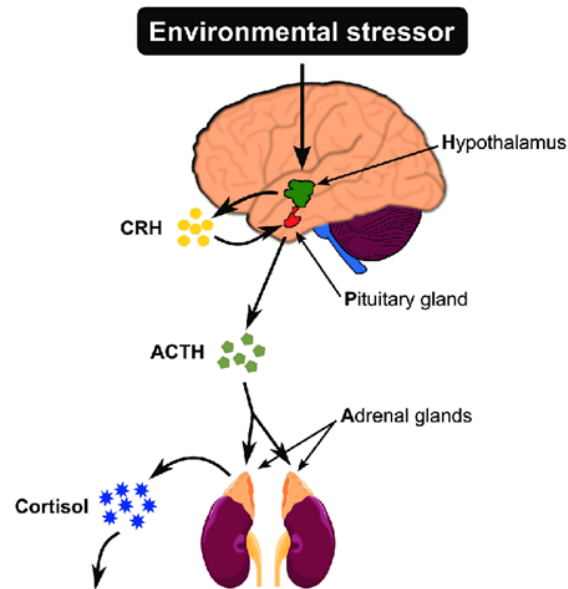


Figure 7: The hypothalamic-pituitary-adrenal (HPA) neuroendocrine axis triggered by an environmental stressor. The cascade details the sequential release of CRH and ACTH, culminating in the adrenal cortex synthesizing cortisol, which serves as a vital target analyte for epidermal stress monitoring patches.

6.2. Compliant Ultrasound Transducer Arrays

Complementing biochemical sensors, thin-film wearable ultrasound patches represent a powerful breakthrough in structural clinical monitoring. Utilizing advanced structural configurations where miniaturized rigid piezoceramic micro-transducer elements are linked together via stretchable serpentine wires and encapsulated in soft silicone matrices, these arrays easily conform to the skin.

These flexible ultrasound probes act as linear or curvilinear scanner systems, operating in direct contact with skin without hindering patient mobility. They measure raw blood motion, vessel diameter transformations, and myocardial wall performance with accuracy matching bulky commercial hospital probes. These systems assist in point-of-care guidance for peripherally inserted central catheter (PICC) needle tracking, continuous abdominal organ scanning, and real-time urinary bladder volume estimations. Furthermore, Transcranial Ultrasound Stimulation (TUS) has emerged as an influential neuromodulatory technique, focusing pulsed ultrasonic waves through the cranium to non-invasively regulate deep neural activity, showing therapeutic promise for Alzheimer's disease, epilepsy, and acute stroke rehabilitation.

7. Closed-Loop AIoT Integration and Future Outlook

A modern personal healthcare architecture requires moving beyond isolated sensor nodes toward a highly integrated Body Area Network (bodyNET). As demonstrated by pioneering closed-loop frameworks, a bodyNET system joins multiple battery-free, ultra-thin biochemical and biophysical patches distributed across the body. These decentralized sensors transfer raw

data via passive Radio-Frequency Identification (RFID) or Near-Field Communication (NFC) links to a flexible read-out master board integrated into everyday clothing.

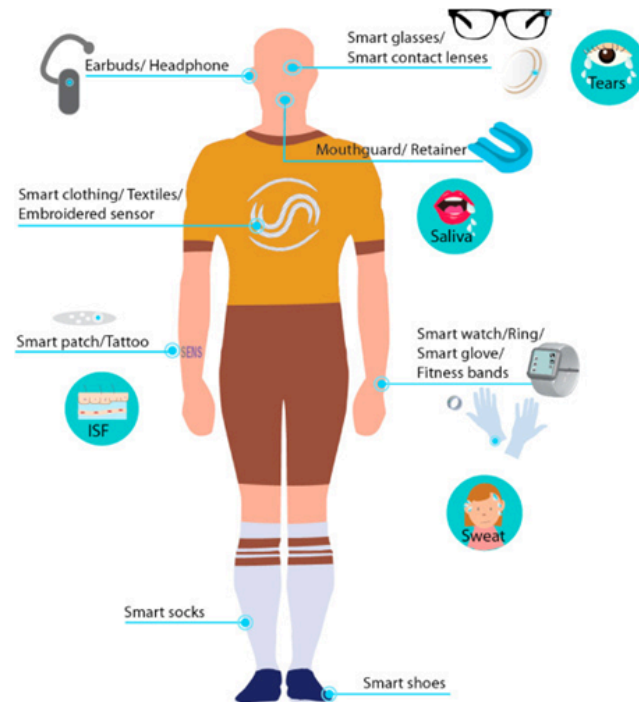


Figure 8: Spatial distribution of diverse wearable form factors across the human body, illustrating custom-tailored nodes including smart contact lenses (tears), mouthguards (saliva), smart wristbands/tattoos (sweat), and smart garments/socks/shoes optimized for real-time biopotential and kinetic capture.

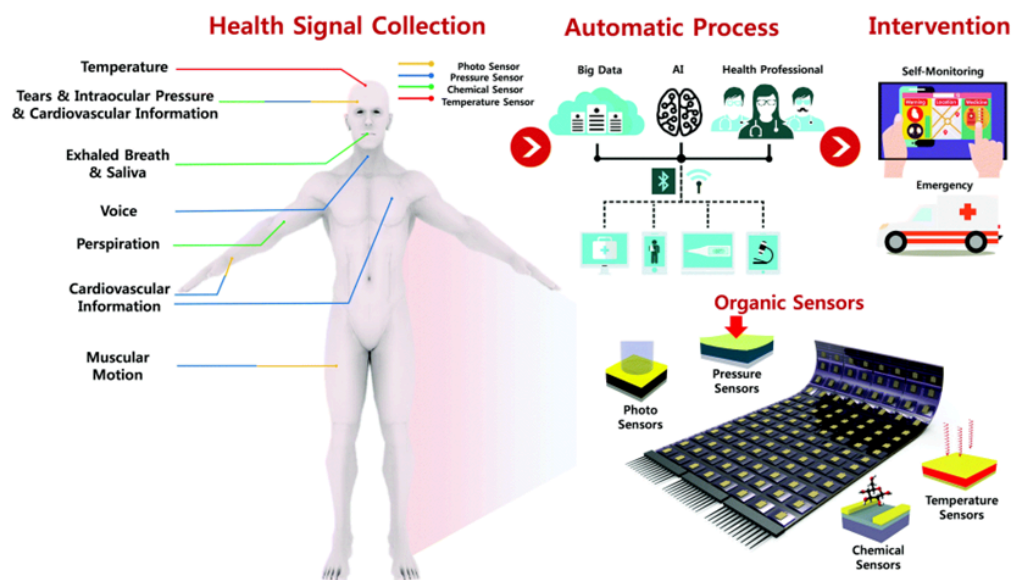


Figure 9: Unified pipeline of an Intelligent Wearable Personal System (IWPS) operating within an AIoT environment. The three-tiered matrix shows: (1) multi-modal Health Signal Collection via organic sensor arrays, (2) Automatic Processing driven by Big Data, cloud computation, and

Machine Learning classification models, and (3) real-time medical Intervention and emergency response.

7.1. Artificial Intelligence and 5G/6G Networks

The integration of emerging Artificial Intelligence (AI) and Machine Learning (ML) algorithms with multi-modal wearable systems marks a turning point in remote clinical care. Utilizing high-speed 5G/6G and wireless networks, the large data streams captured by multi-analyte skin arrays are aggregated into cloud servers.

Machine learning architectures process these expansive training sets to perform multi-analyte calibration. For example, the catalytic activity of oxidase enzymes in sweat sensors is heavily influenced by ambient temperature fluctuations and local sample pH variations. By co-integrating solid-state pH and temperature sensors onto the same platform as the biochemical sensor, an ML model can dynamically calibrate raw sensor currents in real time based on environmental variables, ensuring true quantitative precision. Furthermore, trained neural networks execute complex predictive analytics, analyzing hidden multi-analyte correlations to identify early markers of clinical deterioration, predict future cardiovascular events, detect athletic overtraining fatigue, or automate transdermal drug delivery systems.

7.2. Critical Translation Challenges and Strategic Horizons

While the academic and industrial growth of flexible chemical sensors is substantial, several technical hurdles must be overcome before widespread clinical commercialization:

1. **Biofouling and Surface Degradation:** Bioreceptor surfaces exposed to complex biofluids suffer from surface contamination due to non-specific protein adsorption and cellular debris, causing signal drift and sensitivity loss. Developing anti-fouling zwitterionic polymer coatings and highly stable synthetic receptors (e.g., molecularly imprinted polymers [MIPs]) remains an active area of research.
2. **Autonomous Power Infrastructure:** Conventional lithium-ion batteries are heavy, rigid, and have a limited lifespan, creating electronic waste challenges. Next-generation research focuses on self-powered systems driven by flexible triboelectric, thermoelectric, and photovoltaic energy harvesters that convert body motion, sweat chemistry, or ambient light into electrical power.
3. **Safety and Nanotoxicity:** The inclusion of nanomaterials, such as certain carbon nanotubes and quantum dots, introduces potential health risks if they detach from the substrate matrix and penetrate the epidermal barrier. Ensuring rigorous encapsulation, using entirely non-toxic organic polymers, and establishing clear regulatory safety frameworks are vital for clinical translation.

8. Conclusion

Flexible chemical sensors and biosensors represent a fundamental tool for future healthcare ecosystems. By merging advanced materials science, microfluidic handling, and machine

learning models, these systems are poised to shift medicine from a reactive model to a proactive, continuous, and deeply personalized paradigm.

9. Reference

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